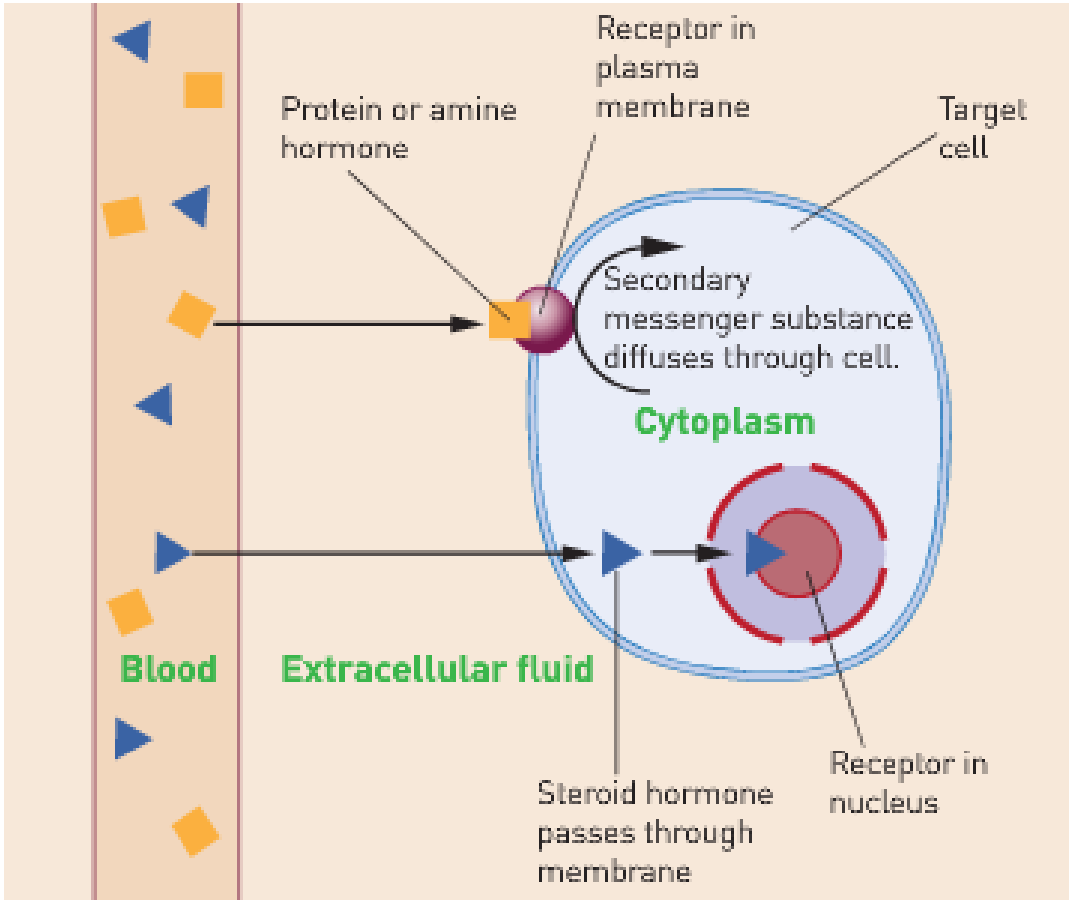
Hormone Action at the Cellular Level:
Protein and Amine Hormones



Entry to target cells depends on hormone type

- Some hormones are made from proteins, or protein components such as amines and peptides
- Other hormones are made from steroids
- Their chemical structure affects how they have their action on the target cell:

  - Steroid hormones can pass through the cell membrane as they are lipid soluble.
  - Protein/Amine hormones cannot pass through the cell membrane – they are water soluble.

Protein / Amine / Peptide Hormones

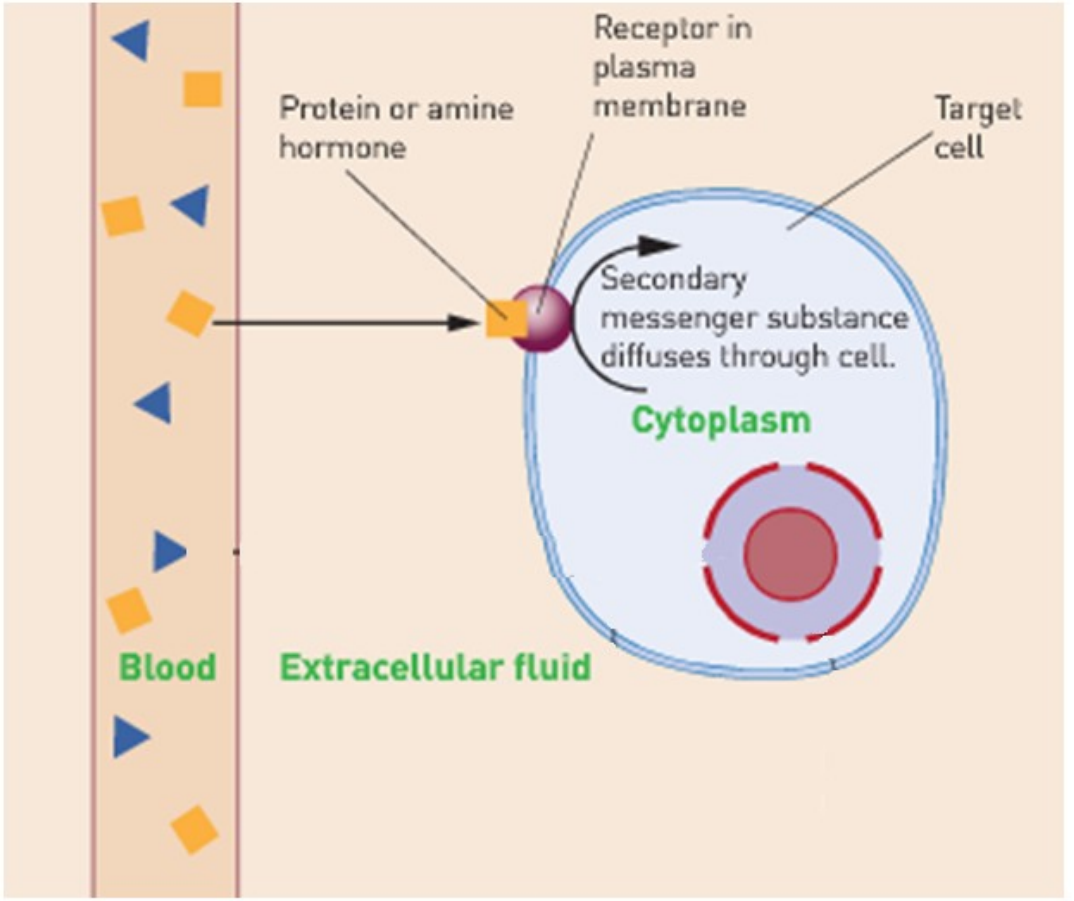
These are water soluble, so they can't pass through the cell membrane.

1: Hormone attaches to specific receptor on cell surface.

2: Hormone/receptor complex triggers release of a second messenger.

3: Second messenger diffuses through the cell and activates and amplifies enzymes that change cell functioning.

Note: the second messenger is broken down quickly, so duration of hormone action is **shorter** than for steroid hormones.



Learning Aim: Describe how protein/amine/peptide hormones function in the cell.

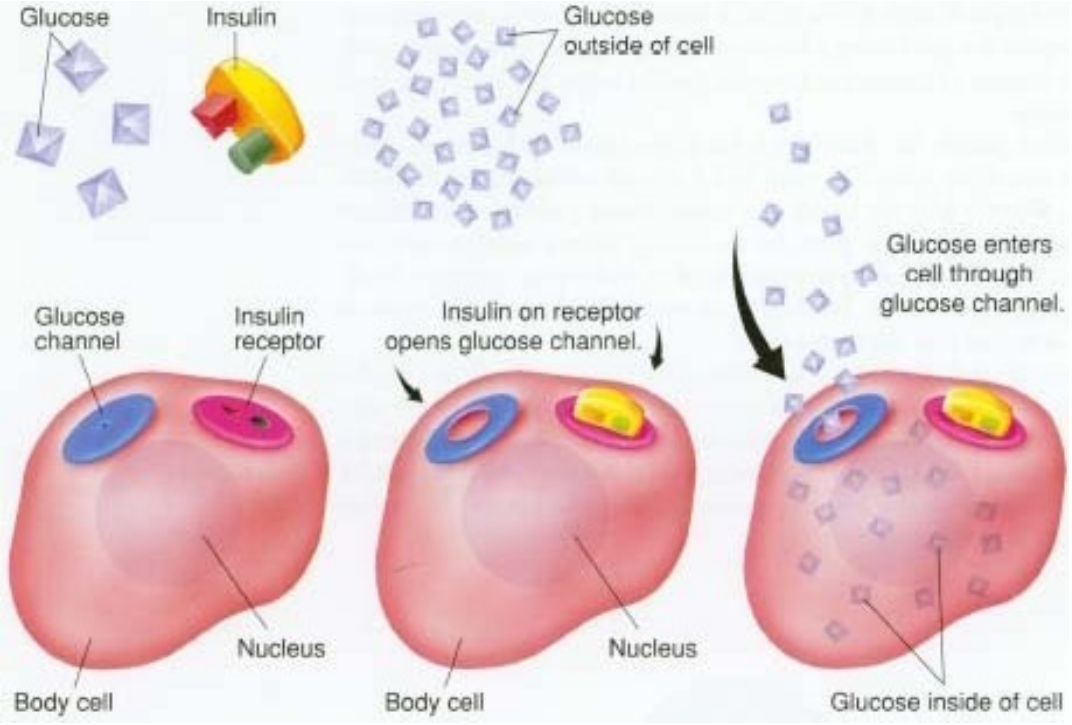
Enzyme amplification

- For some hormones, each hormone molecule activates thousands of enzyme molecules
- This is one way that a second messenger can work for protein and amine hormones
- Occurs due to a process called *enzyme amplification:*
  - Hormone triggers a cascade of chemical reactions where the number of molecules involved increases exponentially at each step.
  - One hormone molecule can trigger production of billions of enzyme molecules

Learning Aim: Briefly describe enzyme amplification

Protein and amine hormone examples

- Example: Insulin
- produced by pancreas
- enters bloodstream
- unlocks cells so glucose can enter.
- some tissues have more receptors than others



Learning Aim: Give examples of protein/amine/peptide hormones

Other protein/amine hormone examples

- Growth hormone (GH)
- Prolactin (PRL)
- Thyroid-stimulating hormone (TSH)
- Adrenocorticotropic hormone (ACTH)
- Follicle-stimulating hormone (FSH)
- Luteinizing hormone (LH)
- Antidiuretic hormone (ADH)
- Oxytocin
- Thyroxine (T_4), triiodothyronine (T_3)
- Calcitonin
- Parathyroid hormone (PTH)
- Epinephrine, norepinephrine
- Melatonin
- Insulin
- Glucagon

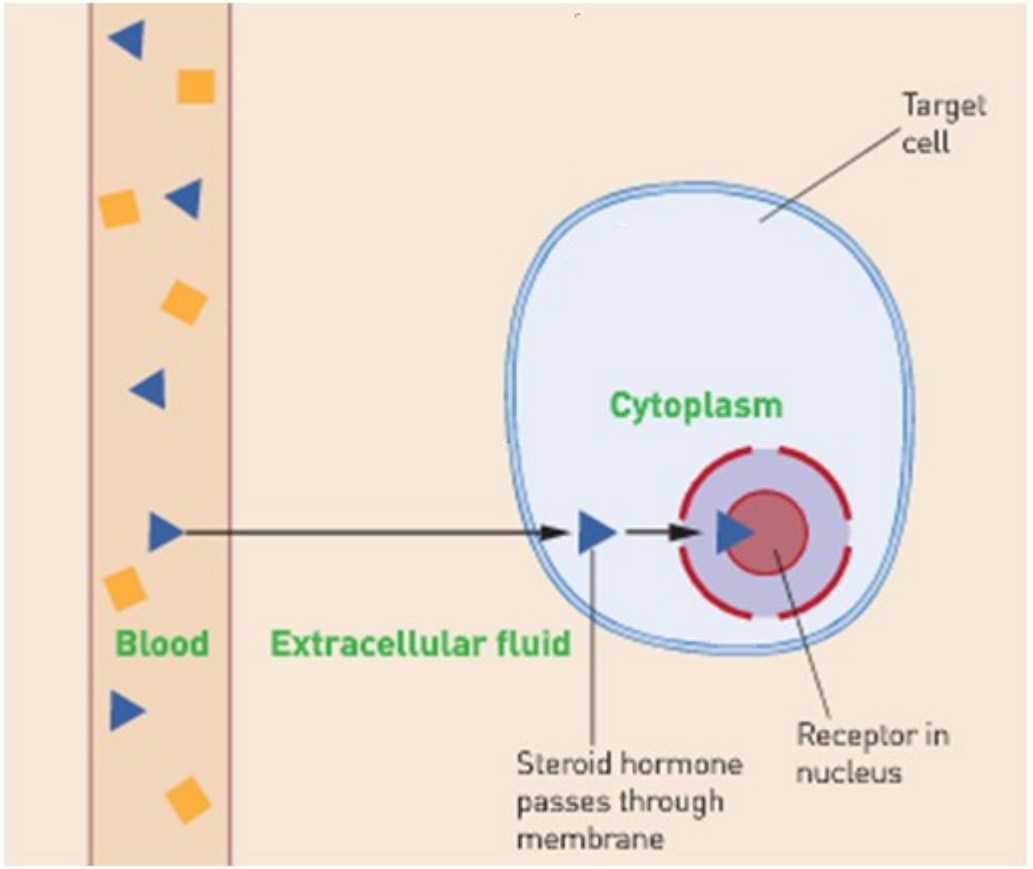
Steroid hormones

1: Hormone passes through cell membrane as they are lipid soluble.

2: Hormone binds to receptors *inside* cell.

- Eg in nucleus, on mitochondria or other organelles

3: Hormone activates genes to change type, activities or quantity of protein produced:



- May:
  - Activate genes to produce an enzyme or structural protein
  - Change shape of enzyme to turn it on or off
  - Change rate of transcription and translation to control rate of protein production
- Hormone takes longer to clear, so **duration of action is longer** than for protein/amine hormones

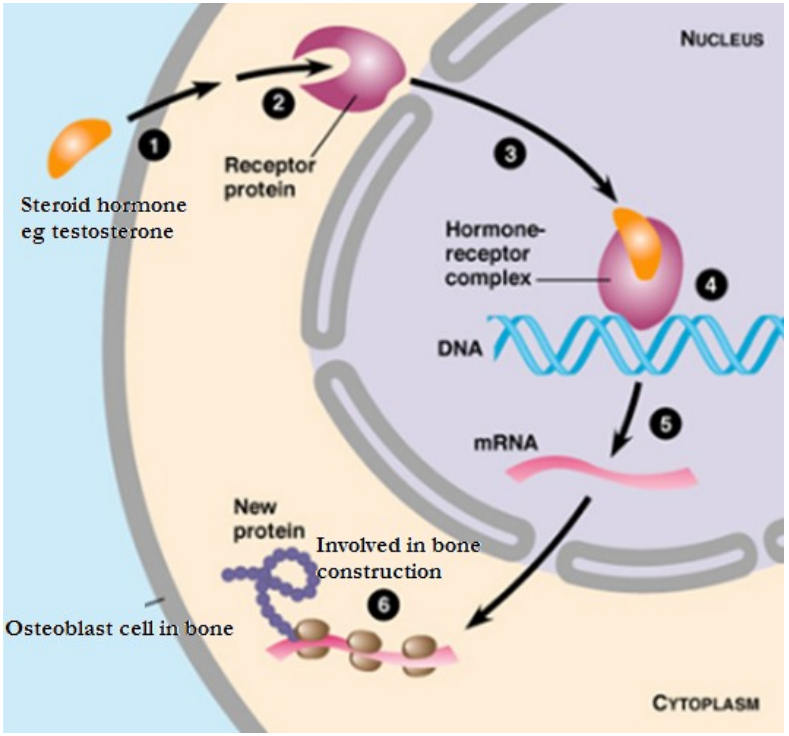
Learning Aim: Describe how steroid hormones function in the cell.

Steroid hormone examples

- Eg: Testosterone

Testosterone passes through the cell membrane, and binds to a receptor protein within a bone cell, which then moves into the nucleus and activates genes which code for the production of proteins (eg collagen) involved in bone construction.

Note: testosterone has different effects in a number of different body cells.



Other steroid hormone examples:

Aldosterone

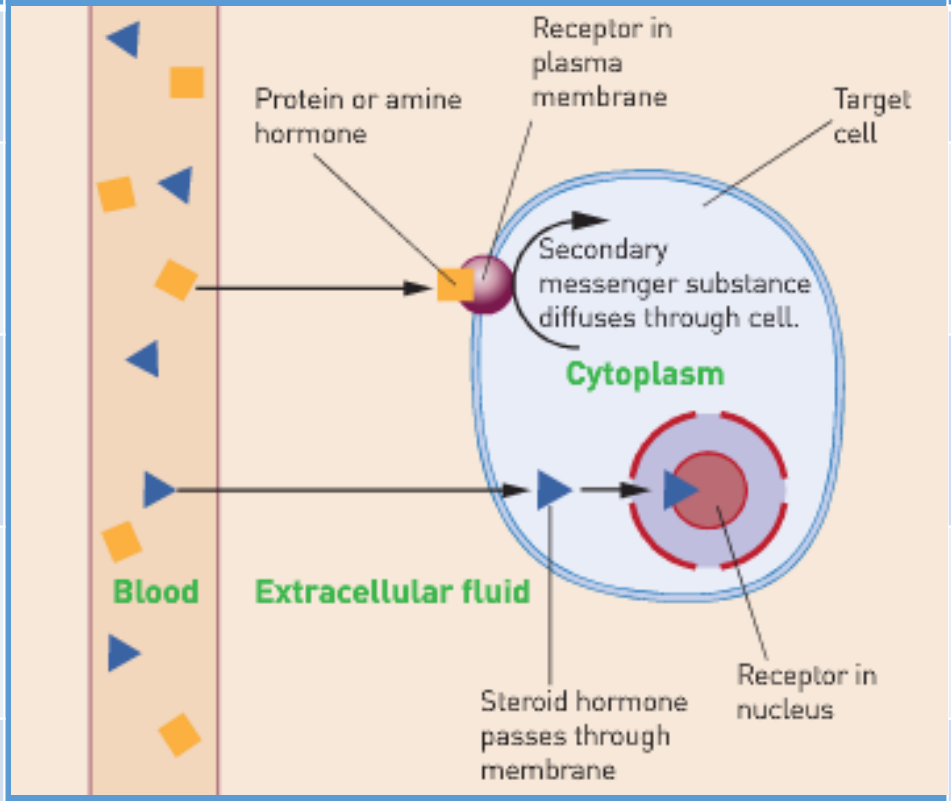
Cortisol, corticosterone, cortisone

Testosterone

Estrogens and progesterone

Learning Aim: Give examples of steroid hormones

Comparing Protein/Amine and Steroid Hormones

	Protein/Amine hormones		Steroid Hormones
Receptor site	On cell membrane		Inside cell on nucleus or organelles
Action of hormone/receptor complex	Activates second messenger substance		Hormone receptor complex moves to nucleus
Effect on cell	Second messenger activates enzymes		activates genes to change type, activities or quantity of protein produced
Duration of action	Shorter – second messenger broken down quickly.		Longer – hormone broken down more slowly
Examples	Thyroxine, Growth Hormone and many others		Testosterone, Oestrogen, Cortisol, and a few others